

The Macroeconomic Effects of Climate Policy Uncertainty

Konstantinos Gavriilidis
University of Stirling

Diego R. Känzig
Northwestern University

Ramya Raghavan
Northwestern University

James H. Stock
Harvard University

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Motivation

Uncertain Times for Climate Policy

Paris climate accord to take effect; Obama hails 'historic day'

By Alister Doyle and Roberta Rampton

October 5, 2016 5:03 PM PDT · Updated October 5, 2016



U.S. President Barack Obama delivers a statement on the Paris Agreement in the Rose Garden of the White House in Washington, U.S., October 5, 2016. REUTERS/Yuri Cripas [Purchase Licensing Rights](#)

- U.S. climate policy marked by reversals and uncertainty

- Inconsistent stance on Paris agreement

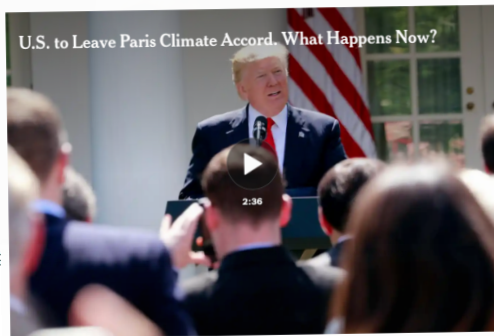
Uncertain Times for Climate Policy

Paris climate accord to take effect;

Trump Will Withdraw U.S. From Paris Climate Agreement

By Michael D. Shear

June 1, 2017



Brad Plumer, a climate reporter for The New York Times, explains the consequences of President Trump's decision today that he will withdraw from the landmark global warming agreement. Doug Mills/The New York Times

- U.S. climate policy marked by reversals and uncertainty
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Uncertain Times for Climate Policy

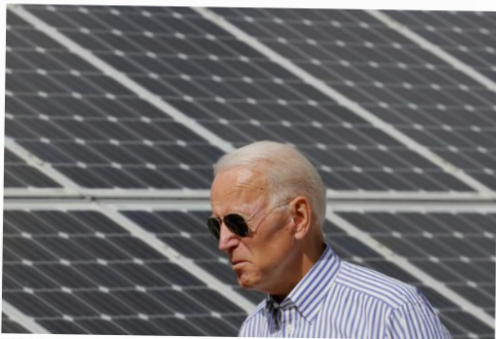
Paris climate accord to take effect;

Trump Will Withdraw U.S. From

***Biden announces return to global climate
accord, new curbs on U.S. oil industry***

By Valerie Volcovici and Trevor Hunnicutt

January 20, 2021 6:46 PM PST · Updated January 20, 2021



FILE PHOTO: Democratic 2020 U.S. presidential candidate and former Vice President Joe Biden walks past solar panels while touring the Plymouth Area Renewable Energy Initiative in Plymouth, New Hampshire, U.S., June 4, 2019. REUTERS/Brian Snyder/File Photo [Purchase Licensing Rights](#)

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Biden announces return to global climate

Trump is withdrawing from the Paris Agreement (again), reversing U.S. climate policy

JANUARY 21, 2025 · 5:00 AM ET

By Nate Perez, Rachel Waldholz



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- Inconsistent stance on Paris agreement
- Many other examples ...

Uncertain Times for Climate Policy

(a) Timeline of the U.S. stance on the Paris Climate Agreement



(b) Timeline of the Waxman-Markey bill



(c) Timeline of California's waiver to set stricter emissions standards



Uncertain Times for Climate Policy

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- U.S. climate policy marked by reversals and uncertainty
 - Inconsistent stance on Paris agreement
 - Many other examples ...
- Salient dimension of policy uncertainty
 - Makes it difficult for firms and households to plan
- How does climate policy uncertainty affect the economy?

This Paper

- Construct a new measure of **climate policy uncertainty** (CPU) based on *newspaper coverage* in the United States
 - Building on approach by Baker, Bloom, and Davis (2016)
 - Index spikes near important events related to climate policy
Presidential announcements on international climate agreements, congressional debates on climate bills, or disputes about the right of the EPA and states to regulate emissions ...
- **Identification challenge**: CPU may increase in response to economic downturns
 - Propose new IV approach: Isolate plausibly exogenous increases in CPU, quantified using newspaper coverage in tight window around events
- Provide new estimates on the **dynamic causal effects** of CPU

Results: Aggregate Impacts

- **Climate policy uncertainty** has significant **macroeconomic effects**
 - Higher uncertainty causes fall in **output**, private **investment**, and **employment**
 - But also **increases commodity** and **consumer prices**
 - Emissions fall following economic contraction, **no green paradox** at aggregate level
- Climate policy uncertainty transmits to the economy as **supply shocks**
 - Unlike **economic policy uncertainty**, which moves output and prices together
 - Important implications for monetary policy
- **No increase** in other measures of uncertainty
 - Climate policy uncertainty is a distinct source of policy uncertainty
- **No effect** on government spending and emissions intensity unchanged
 - Successfully capture uncertainty and not news

Results: Firm-Level Effects

- **Climate policy uncertainty** has substantial **firm-level impacts**
 - Firms view climate policy uncertainty as material financial risk
- Firms respond more strongly when their **climate change exposure** is high
- Rich **sectoral heterogeneity** in firm-level responses
 - Most sectors lower investment and R&D
 - Mining, oil, and utilities **increase investment**
- **But:** R&D decreases particularly strongly in these sectors
 - **Green paradox** at micro level
 - Exacerbate transition costs through misallocative forces

Measuring Climate Policy Uncertainty

Defining Climate Policy Uncertainty

- **Definition:** Lack of clarity/predictability of government actions on climate change
 - Focus on climate policy with **national significance**
 - Includes uncertainty about **new climate policies** as well as political/legal **challenges to existing policies**
- **Measurement** is challenging:
 - Follow approach by Baker, Bloom, and Davis (2016) leveraging informational content in **newspaper articles**

Measuring Climate Policy Uncertainty

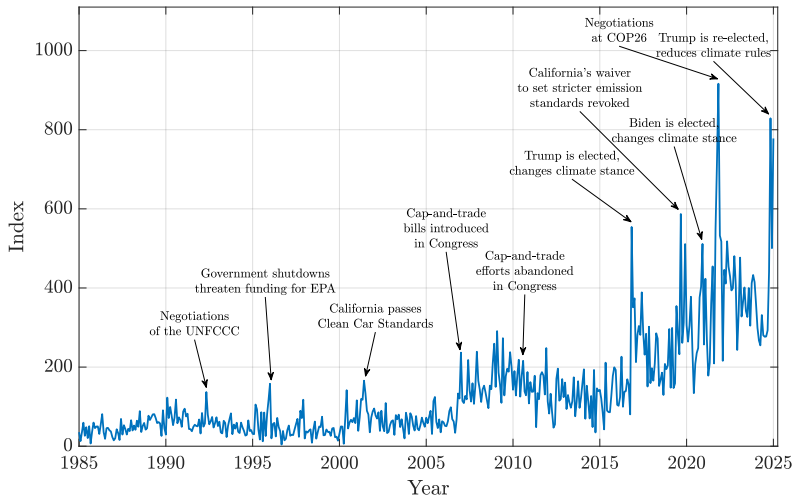
- **Idea:** Use dictionary of words whose occurrence in newspaper articles is associated with coverage of topics related to CPU
 - **Climate:** *climate change, carbon dioxide, greenhouse gas, clean energy, ...*
 - **Policy:** *regulation, legislation, white house, congress, ...*
 - **Climate policy:** *carbon tax, emissions trading, EPA, climate bill, ...*
 - **Uncertainty:** *uncertain**
- **Dictionary Source:** Corpus of news articles from specialized climate policy reporting agencies:
Inside Climate News, Inside EPA, Washington Week (Energy)
- **Selection:** Identify article as CPU if it contains at least one term in:
(**Climate** AND **Policy** AND Uncertainty) OR (**Climate policy** AND Uncertainty)

Measuring Climate Policy Uncertainty

- **Sample:** ~ 7.87 million news articles published in leading American newspapers from mid-1980s
 - *New York Times, Wall Street Journal, Washington Post, LA Times*
 - Comprehensive and systematic coverage of national climate policy developments
- **Construction:** Index counts, each month, the number of articles discussing uncertainty about climate policy, divided by the total number of published articles
 - Manual and LLM review of article samples reveals few false positives
 - Results robust to varying dictionary terms, expanding set of newspapers, and normalization

Climate Policy Uncertainty Since the 1980s

Figure 3: Climate Policy Uncertainty Index



Climate Policy Uncertainty Since the 1980s

- **Climate policy uncertainty increased** substantially, especially in recent years
 - Some notable spikes in first part of the sample, marked increase in late 2000s amid emissions trading proposals
 - Stark increase in uncertainty following Paris agreement and election of Trump
- Index uncorrelated with VIX and geopolitical risk
- Index weakly correlated with EPU and trade policy uncertainty
 - Results robust to controlling for other uncertainty measures
 - CPU captures **distinct variation** from other dimensions of policy uncertainty

Identification

Identification

- Uncertainty about climate policy may increase in times of **economic distress**
- Isolate plausibly **exogenous** increases in climate policy uncertainty
 - Driven by climate-related, political, or ideological considerations
- Based on **narrative account** of U.S. climate policy history, identify 146 events
 - Legislative, regulatory, & judicial actions leading to climate policy uncertainty [▶ More](#)
- **Examples:**
 - Inconsistent stance on international agreements like Kyoto or Paris agreement
 - Debates over proposed legislation such as cap-and-trade policies
 - Disputes about the right of the EPA and states to regulate emissions

IV for Climate Policy Uncertainty

- Events may contain both **policy news** and uncertainty. But:
 - Changes in policy stringency and uncertainty often move in opposite directions
 - Direction of stringency is readily observed
- Our approach thus consists of **two steps**:
 1. Measure change in **climate policy reporting intensity** around events d :

$$\Delta cp_{i,d}^{\text{intensity}} = n_{i,d}^{\text{cp}} - n_{i,d-1}^{\text{cp}}$$

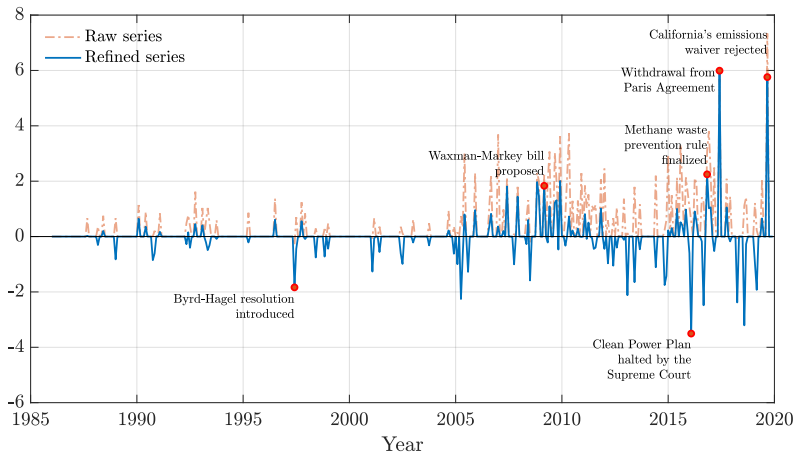
2. Purge reporting intensity from changes in **climate policy stringency**

$$\Delta cp_{i,d}^{\text{intensity}} = \alpha_i + \beta_i \times \Delta cp_{i,d}^{\text{stringency}} + \Delta cp_{i,d}^{\text{uncertainty}}$$

- Aggregate over newspapers and time to **monthly IV** series, $\Delta cp_t^{\text{uncertainty}}$

Major Climate Policy Uncertainty Events

Figure 4: Climate Policy Uncertainty Event Series



Econometric Framework

- Use $z_t = \Delta \text{cp}_t^{\text{uncertainty}}$ as IV to identify a **climate policy uncertainty shock**

- **Identifying assumptions:**

► Details

$$\mathbb{E}[z_t \varepsilon_{1,t}] = \alpha \neq 0$$

(Relevance)

$$\mathbb{E}[z_t \varepsilon_{2:n,t}] = \mathbf{0},$$

(Exogeneity)

- **Estimation:** We rely on VAR techniques:

- Sample: 1985 – 2019
- Specification: 6 variables, 12 lags

CPU index, industrial production, unemployment rate, commodity prices, consumer prices, policy rate

► Data

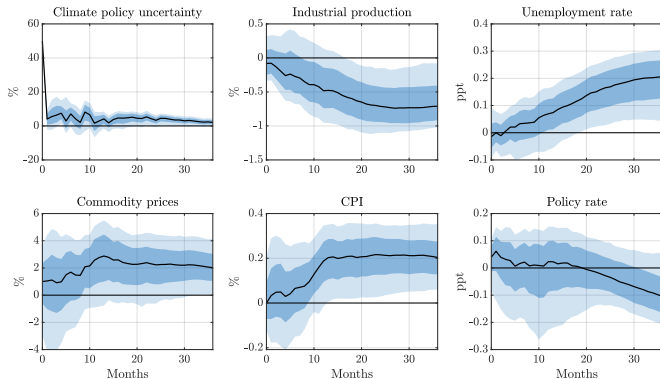
- Use local projections as robustness and to map out wider effects

► More

Aggregate Effects

Macroeconomic Effects of Climate Policy Uncertainty

Figure 5: Baseline VAR

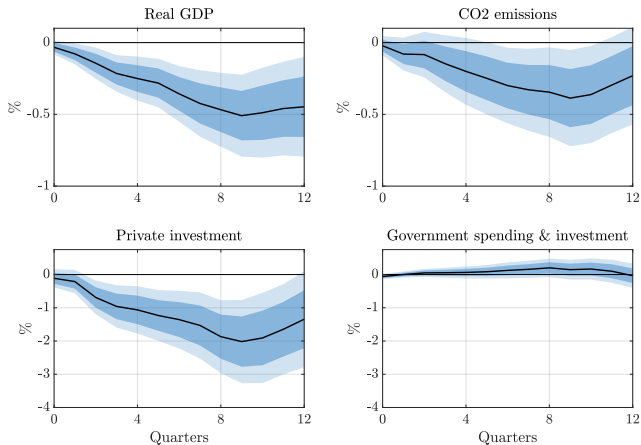


First-stage regression: F-statistic: 23.54, R^2 : 2.81%

- Significant fall in **industrial production** and rise in **unemployment**
- **Commodity** and **consumer prices** increase
- Monetary response is ambiguous
- Transmit more like **supply shocks**

Macroeconomic Effects of Climate Policy Uncertainty

Figure 6: Impacts on GDP, Emissions and Investment



- Significant fall in **GDP** and **investment**
- No response of **government spending** and **investment**
- **Emissions** fall but emissions intensity unchanged
 - No evidence for **green paradox**

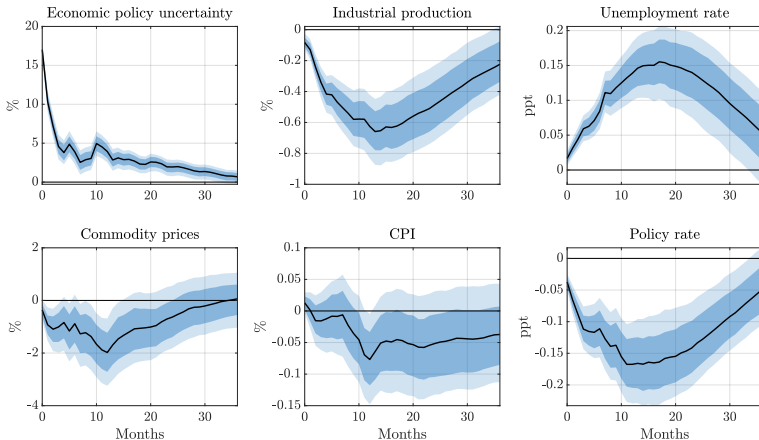
- CPU has no significant effect on **other uncertainty measures** [▶ Detail](#)
 - Economic policy uncertainty, trade policy uncertainty, geopolitical risk, financial uncertainty, ...
- Results robust to:
 - **Measurement** of climate policy uncertainty [▶ Detail](#)
 - **Event selection** and **instrument construction** [▶ Detail](#)
 - Controlling for **other uncertainty measures** [▶ Detail](#)
 - Controlling for first moment using **climate news** and **climate sentiment** index [▶ Detail](#)
 - Additional controls (physical climate, political environment) [▶ Detail](#)
 - Relaxing VAR **assumptions** (invertibility, dynamic VAR structure) [▶ Detail](#)
 - VAR **specification** [▶ Detail](#)

Is Climate Policy Uncertainty Special?

- What do we learn from looking at **climate policy uncertainty**?
- Contrast with effects of broader **economic policy uncertainty**
 - Use index from Baker, Bloom, and Davis (2016)
 - Estimate responses based on recursive VAR approach

Effects of Economic Policy Uncertainty

Figure 7: VAR with Economic Policy Uncertainty



Climate versus Economic Policy Uncertainty

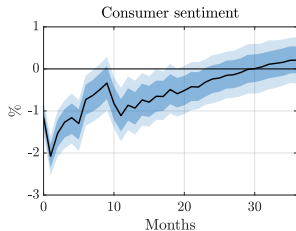
- Economic policy uncertainty transmits **differently** from climate policy uncertainty
 - Economic policy uncertainty leads to fall in production
 - But commodity and consumer prices also tend to **decrease**
 - Monetary response **accommodates** the shock
- They thus transmit more like **demand shocks**
 - True for most uncertainty measures, e.g. an innovation to the VIX has very similar effects
- Response of prices to uncertainty shocks theoretically ambiguous
 - **Different channels:** Precautionary demand, real options, precautionary pricing, ...
 - Price response depends on relative strength of supply- and demand-side effects

Why is Climate Policy Uncertainty Inflationary?

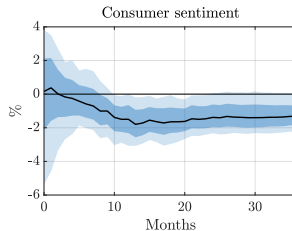
- For CPU **supply-side** effects dominate, for EPU **demand-side** effects dominate
 - Consumer sentiment falls significantly on impact for EPU but **not** for CPU shock

Figure 8: Impact on Sentiment

(a) Economic policy uncertainty shock



(b) Climate policy uncertainty shock



Firm-Level Effects

Firm-Level Impacts

- Revisit effects of **climate policy uncertainty shock** in panel of firms
 - Construct panel of U.S. listed firms
 - Quarterly sample from 1986 to 2019 (136 quarters) with 11,872 firms
- Average effects on sales, employees consistent with aggregate data
- However, average effect may mask substantial **heterogeneity**

► Details

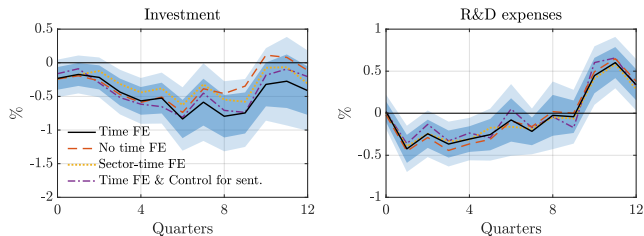
Heterogeneity by Climate Exposure

- How do effects vary with firm-level **climate change exposure**?
 - Use exposure measures by Sautner et al. (2023) based on earnings conference calls
- Estimate local projection on shock **interacted** with exposure

$$y_{i,t+h} = \mu_{i,h} + \delta_t + \theta_h(\text{Exp}_{i,t-1} - \overline{\text{Exp}_i}) \times \varepsilon_{1,t} + \gamma_h' \mathbf{x}_{i,t-1} + \nu_{i,t+h},$$

- Focus on **within-firm** variation to net out permanent differences:
 - How does time- t exposure compare to average exposure of firm i ?
- Allows to control for **time fixed effects**

Figure 9: Heterogeneous Effects Based on Prior Climate Exposure



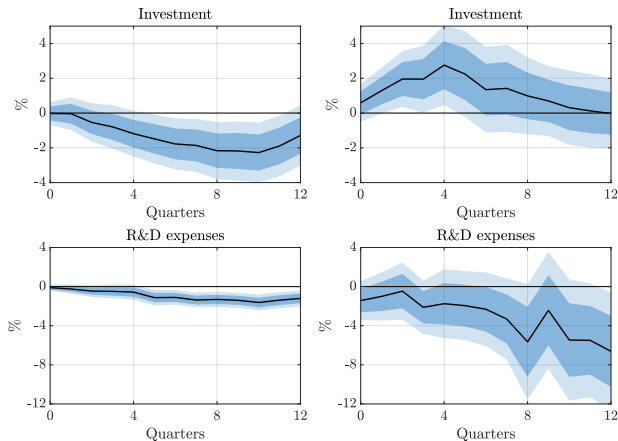
- Firms display **stronger fall** in investment and R&D when **climate exposure** is high
- Robust to time or sector-by-time fixed effects

Sectoral Impacts

Figure 10: Heterogeneous Effects by Sector

(a) All ex. oil, gas, utilities

(b) Oil, gas, and utilities



- How do effects vary by **sectors**?
- Estimate panel LPs, conditioning on different industries
- Most sectors show significant fall in investment and R&D
 - Consistent with **average** response
- **Oil, gas, and utilities** stand out
 - Investment increases
 - R&D falls substantially

► Sales

Longer-Term Impacts

- Results are consistent with **green paradox** at the micro level
- Climate policy uncertainty strengthens incentives to extract fossil fuels
- **But:** Reduces R&D expenses that spur the green transition
- Climate policy uncertainty can exacerbate transition costs via **misallocative** forces
- Confirmed by significant and persistent fall in **TFP**

► Detail

Conclusion

Conclusion

- Climate policy uncertainty has pervasive economic effects at macro and firm-level
 - Broad-based effects beyond brown sectors
 - Effects more pronounced when exposure to climate is high
- Contrary to other uncertainty shocks, climate policy uncertainty transmits more like supply shocks
 - Persistent impacts dragging on investment and innovation
 - Monetary policy can make matters worse by leaning against inflationary pressures
- Illustrates importance of clear and predictable climate policies and coordination between fiscal and monetary policy

Thank you!

Appendix

Related Literature

- **Climate policy actions and macroeconomic effects**

Martin, De Preux, and Wagner (2014); Andersson (2019); Metcalf (2019); Bernard and Kichian (2021); Burke et al. (2023); Kapfhammer (2023); Bilal and Känzig (2024); Känzig (2025); Colmer et al. (2025); ...

- **Measuring climate and environmental policy uncertainty**

Engle et al. (2020); Gavriilidis (2021); Basaglia et al. (2022); Berestycki et al. (2022); Noailly, Nowzohour, and Van Den Heuvel (2022); Sautner et al. (2023); Palikhe, Schaur, and Sims (2024); Basaglia et al. (2025); Marotta, Pagliari, and de Winter (2025); ...

- **Methodology: Uncertainty shocks and narrative identification**

Saiz and Simonsohn (2013); Bloom (2014); Baker, Bloom, and Davis (2016); Hassan et al. (2019); Caldara et al. (2020); Caldara and Iacoviello (2022); Gambetti et al. (2023); ...

- **Theory: Uncertainty in climate economics**

[◀ Back](#) Fried, Novan, and Peterman (2021); Barnett, Brock, and Hansen (2022); Barnett et al. (2025); ...

Index Validation

- **Validation exercise:**

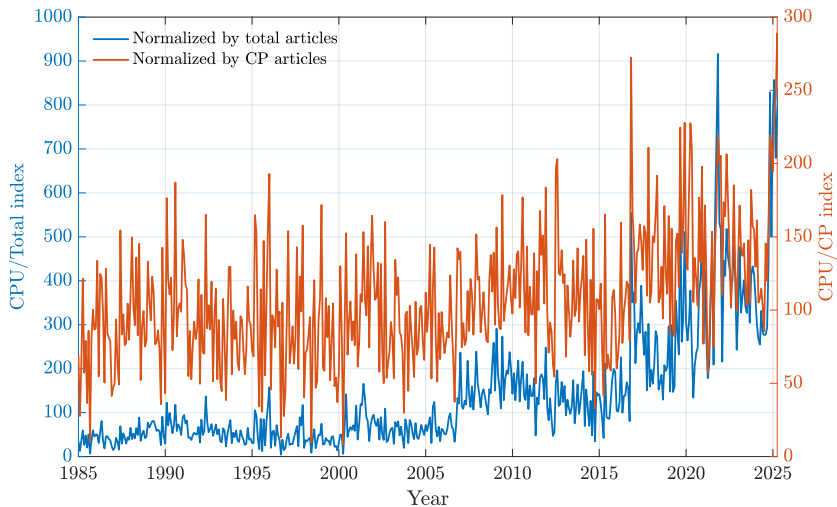
- To validate the index, we use OpenAI's gpt-4o-mini model
- We sample a set of articles from our **Climate** AND **Policy** corpus
- Next, we ask the LLM to classify articles into CPU and non-CPU articles
- This yields a false-positive rate **below 20%**
- Human audit validates LLM classification (~80% concordance)

- **Robustness:**

- Results are robust to using less restrictive set of dictionary terms
- Expanding the set of newspapers

Climate Policy Uncertainty Since the 1980s

Figure 11: Climate Policy Uncertainty Index



Methodology for Event Selection

1. International agreements

- Agreement, signature, or ratification of key treaties and protocols

2. Judicial actions

- Court rulings or stay orders influencing climate policy

3. Legislative actions

- Proposal, introduction, passage, signing, or blocking of climate-related bills
- Pioneering California bills that influence federal policy included

4. Presidential actions

- Statements of intent, policy positions, policy proposals, or executive measures

5. Regulatory actions

- Proposal, final rule, revision, or withdrawal of Federal agency regulations

- **Narrative account:** ✓ **Accords well** with accounts on key historical episodes
- **Forecastability:** ✓ **Not** forecastable by macroeconomic or financial variables
- **Orthogonality:** ✓ **Uncorrelated** with measures of other structural shocks
E.g., other uncertainty, oil, or fiscal shocks

Forecastability: Granger Causality Tests

Variable	p-value
Instrument	0.1191
Climate policy uncertainty	0.7881
Industrial production	0.5294
Unemployment rate	0.7117
Commodity prices	0.1946
PPI	0.9390
CPI	0.7291
Policy rate	0.9874
Climate policy news	0.9118
Climate policy sentiment	0.9980
Economic policy uncertainty	0.4363
Trade policy uncertainty	0.7953
Geopolitical risk	0.6969
VXO	0.2083
Joint	0.9470

Orthogonality: Correlation with Other Shock Measures

Shock	Source	ρ	p-value	n	Sample
<i>Panel A: Uncertainty shocks</i>					
Uncertainty	Bloom (2009)	-0.04	0.48	384	1986M01-2017M12
	Baker, Bloom, and Davis (2016)	0.03	0.54	384	1986M01-2017M12
	Piffer and Podstawski (2017)	-0.02	0.68	355	1986M01-2015M07
<i>Panel B: Oil shocks</i>					
Oil price	Hamilton (2003)	-0.06	0.23	384	1986M01-2017M12
Oil supply	Kilian (2008)	0.04	0.58	225	1986M01-2004M09
	Caldara, Cavallo, and Iacoviello (2019)	0.04	0.48	360	1986M01-2015M12
	Baumeister and Hamilton (2019)	-0.02	0.74	408	1986M01-2019M12
	Kilian (2009)	0.04	0.49	264	1986M01-2007M12
Global demand	Kilian (2009)	-0.07	0.25	264	1986M01-2007M12
Oil-specific demand	Kilian (2009)	0.03	0.63	264	1986M01-2007M12
Oil supply news	Känzig (2021)	0.03	0.53	408	1986M01-2019M12
<i>Panel C: Productivity and news shocks</i>					
Productivity	Basu, Fernald, and Kimball (2006)	-0.03	0.77	104	1986Q1-2011Q4
	Smets and Wouters (2007)	0.10	0.40	76	1986Q1-2004Q4
News	Barsky and Sims (2011)	0.16	0.13	87	1986Q1-2007Q3
	Kurmann and Otrok (2013)	0.14	0.21	78	1986Q1-2005Q2
	Beaudry and Portier (2014)	-0.08	0.42	107	1986Q1-2012Q3
<i>Panel D: Monetary policy</i>					
Monetary policy	Romer and Romer (2004)	0.04	0.66	132	1986M01-1996M12
	Gertler and Karadi (2015)	-0.04	0.50	324	1990M01-2016M12
	Miranda-Agrippino and Ricco (2021)	0.08	0.23	228	1991M01-2009M12
	Bauer and Swanson (2023)	0.04	0.48	383	1988M02-2019M12
	Aruoba and Drechsel (2024)	-0.01	0.81	274	1986M01-2008M10
<i>Panel E: Fiscal policy shocks</i>					
Fiscal policy	Romer and Romer (2010)	-0.04	0.68	88	1986Q1-2007Q4
	Fisher and Peters (2010)	0.00	0.98	92	1986Q1-2008Q4
	Ramey (2011)	-0.07	0.48	100	1986Q1-2010Q4
<i>Panel F: Financial shocks</i>					
EBP	Gilchrist and Zakrajšek (2012)	-0.04	0.46	360	1986M01-2015M12
Loan supply	Bassett et al. (2014)	0.03	0.78	76	1992Q1-2010Q4

External Instrument Approach

- Structural VAR:

$$\mathbf{y}_t = \mathbf{b} + \mathbf{B}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{B}_p \mathbf{y}_{t-p} + \mathbf{S} \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim N(0, \boldsymbol{\Omega})$$

- **External instrument:** Variable z_t *correlated* with the **shock of interest** but *not* with the **other shocks**
- **Identifying assumptions:**

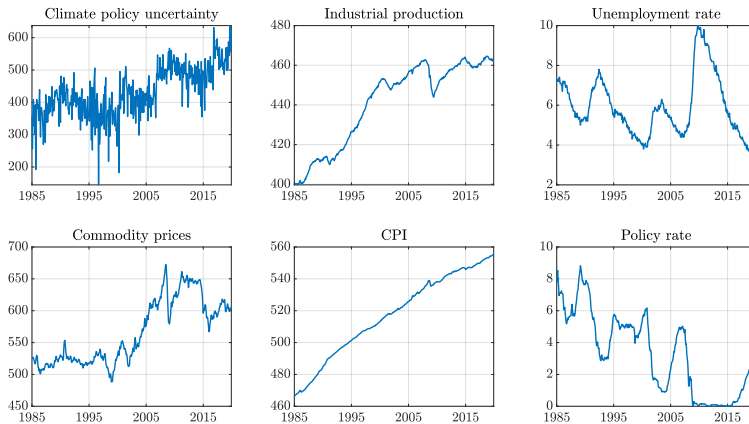
$$\mathbb{E}[z_t \boldsymbol{\varepsilon}_{1,t}] = \alpha \neq 0 \quad (\text{Relevance})$$

$$\mathbb{E}[z_t \boldsymbol{\varepsilon}_{2:n,t}] = \mathbf{0} \quad (\text{Exogeneity})$$

$$\mathbf{u}_t = \mathbf{S} \boldsymbol{\varepsilon}_t \quad (\text{Invertibility})$$

- Use **climate policy uncertainty event series** as an *external instrument* for **climate policy uncertainty index**

Figure 12: Transformed Data Series



Local Projections Approach

1. Estimate impulse responses using local projections-instrumental variables (LP-IV) approach: (Jordà, 2005)

Relax invertibility requirement and dynamic VAR structure

- Use z_t as an instrument for endogenous regressor $y_{1,t}$
- Estimate effects on variables of interest y_i using local projections:

$$y_{i,t+h} = \alpha_h^i + \theta_h^i y_{1,t} + \beta_h^{i'} \mathbf{x}_{t-1} + \nu_{i,t,h},$$

2. Estimate impulse responses to the **climate policy uncertainty shock** from the VAR using local projections: (Stock and Watson, 2018)

Relax dynamic VAR structure to assess possible truncation bias

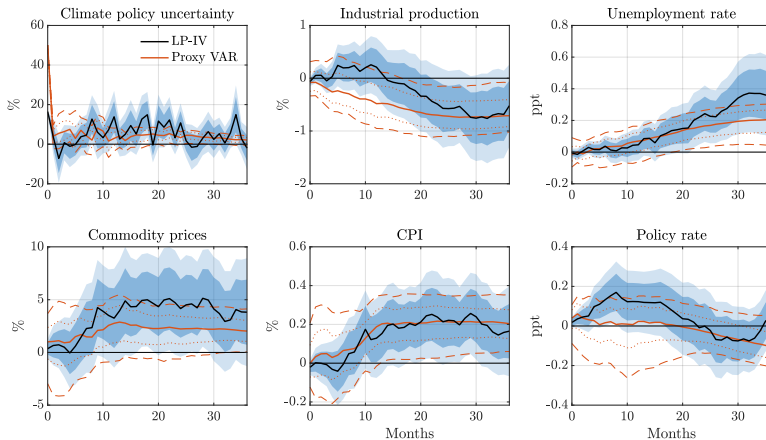
- Extract climate policy uncertainty shock $\varepsilon_{1,t}$ from monthly VAR: $\varepsilon_{1,t} = \mathbf{s}_1' \Sigma^{-1} \mathbf{u}_t$
- Estimate effects on variables of interest y_i using local projections:

$$y_{i,t+h} = \alpha_h^i + \theta_h^i \varepsilon_{1,t} + \beta_{h,1}^i y_{i,t-1} + \dots + \beta_{h,p}^i y_{i,t-p} + \nu_{i,t,h}$$

- Can also estimate effects on variables only available at lower frequencies

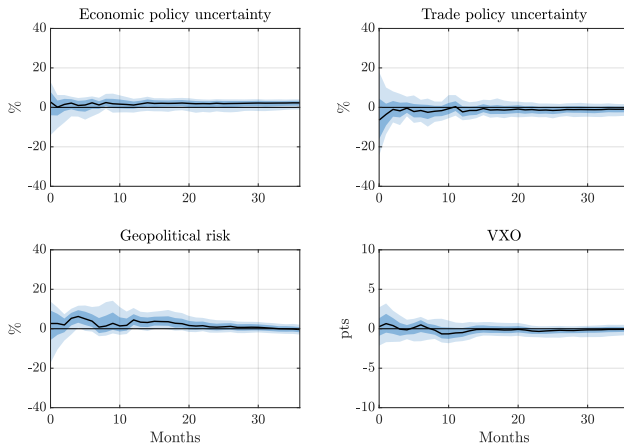
Macroeconomic Effects of Climate Policy Uncertainty

Figure 13: Local Projections



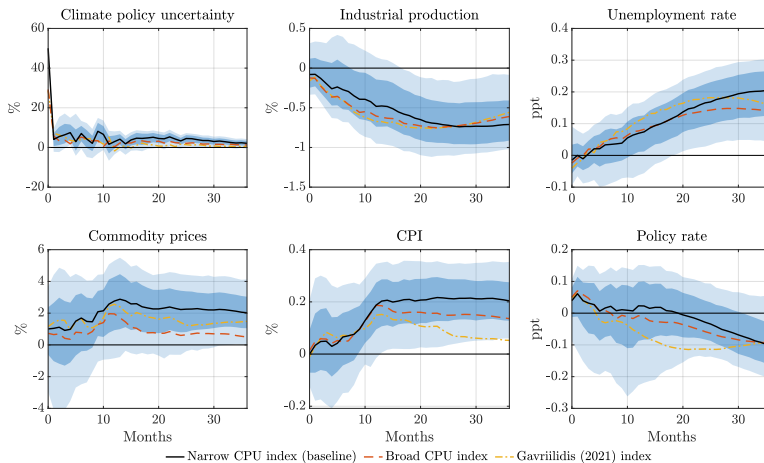
Impacts on Other Uncertainty Measures

Figure 14: Impacts on Other Uncertainty Measures



Construction of the CPU Index

Figure 15: Alternative CPU Measures



Normalization of the CPU Index

Figure 16: Alternative CPU Measures

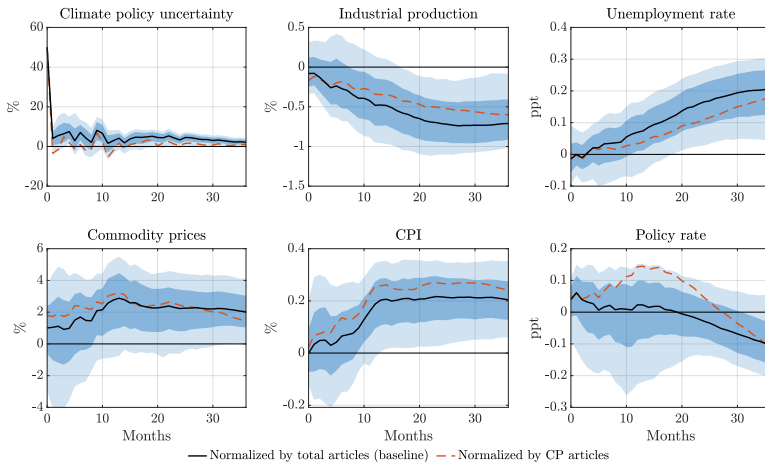


Figure 17: Event Selection

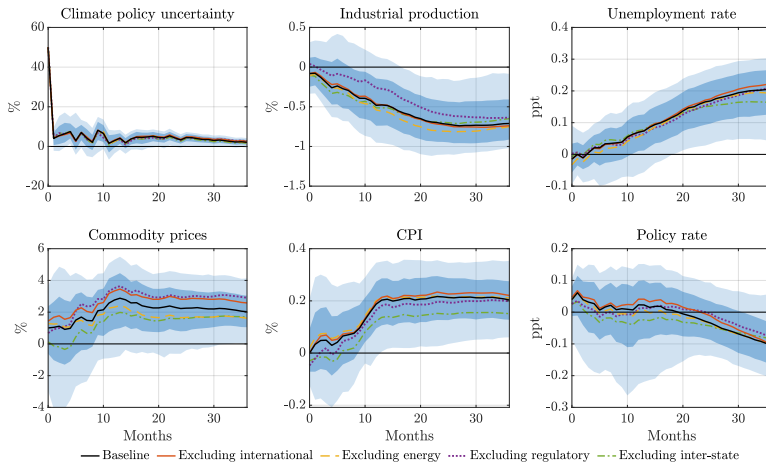


Figure 18: Jackknife Exercise

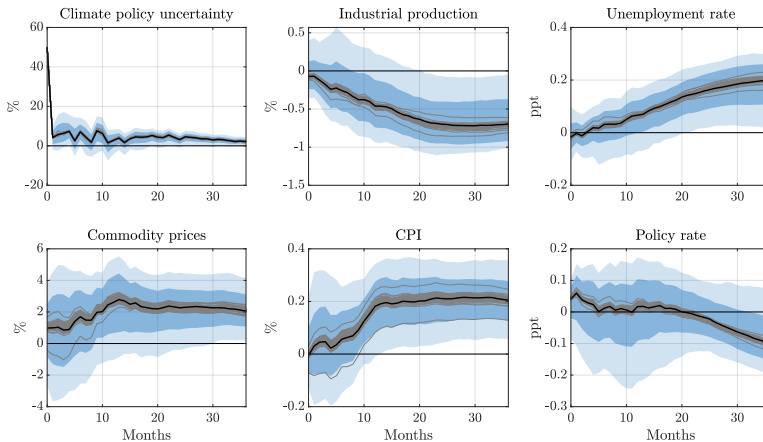


Figure 19: Stringency Index Construction

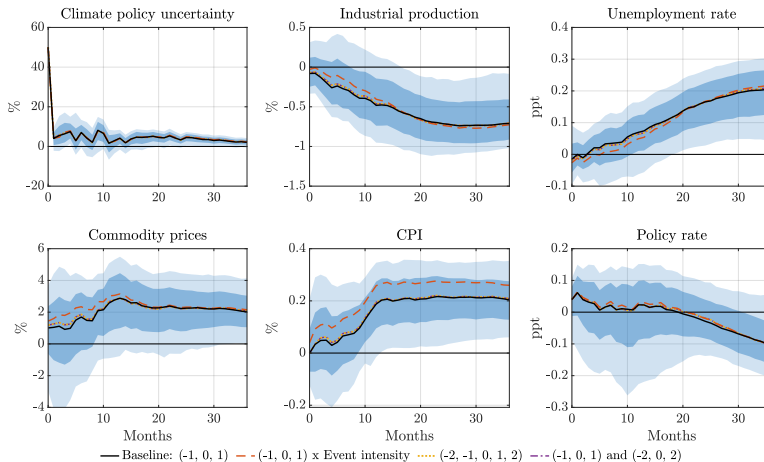


Figure 20: Anticipation and Policy Stringency Controls

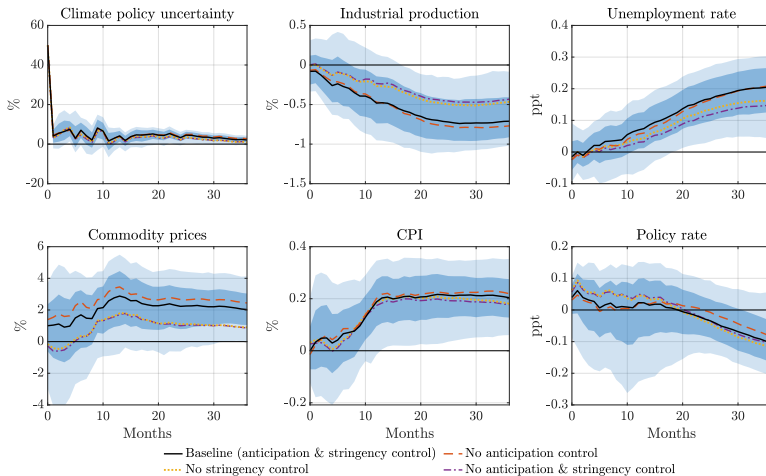
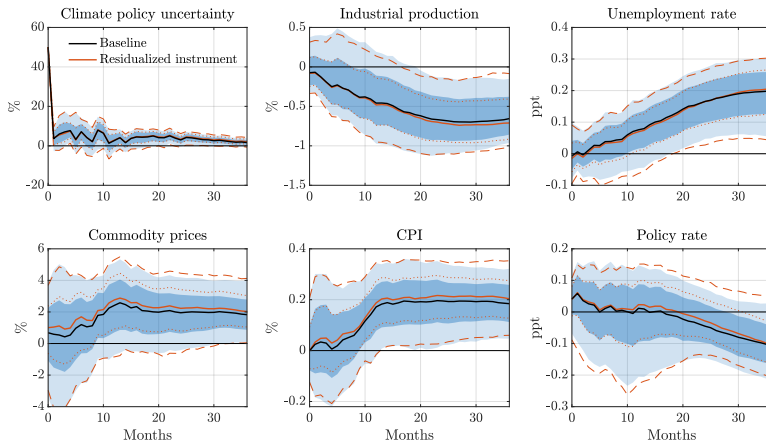


Figure 21: Residualized Instrument



Controlling for News and Other Uncertainty Measures

Figure 22: Additional Controls

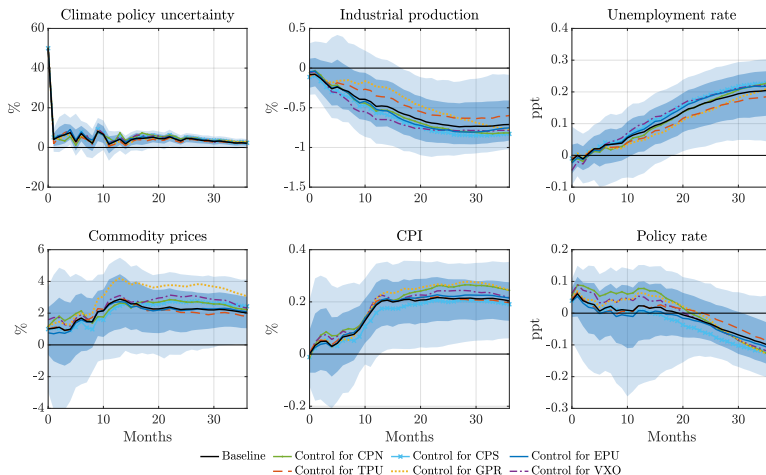


Figure 23: Additional Controls

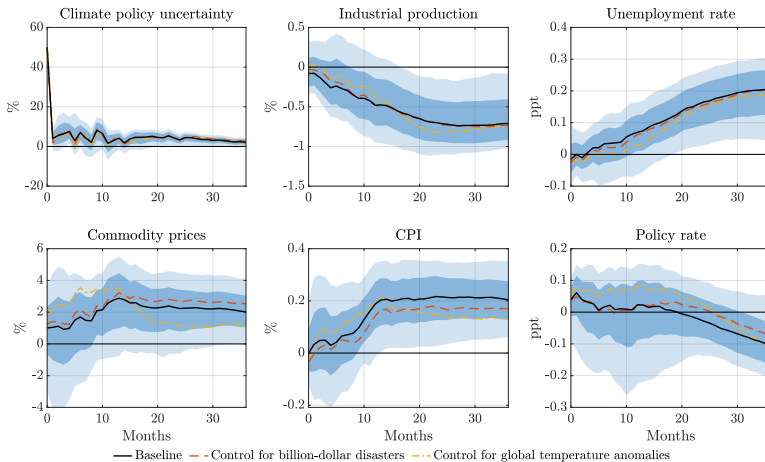
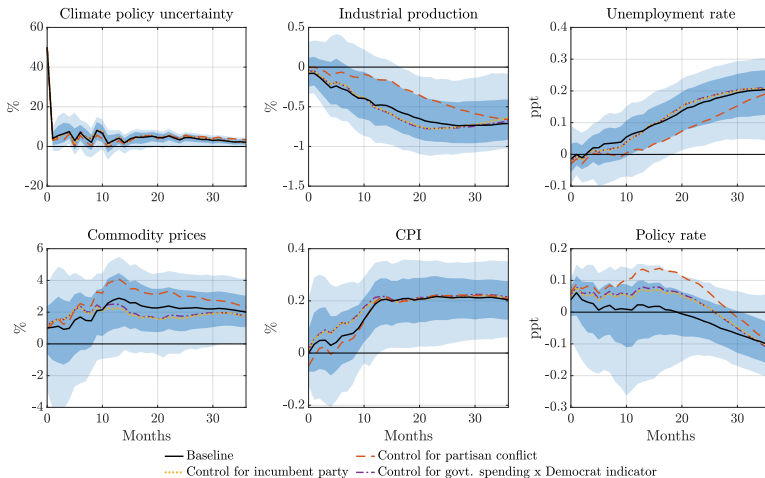


Figure 24: Additional Controls



Relaxing VAR Assumptions

Figure 25: Relaxing VAR assumptions

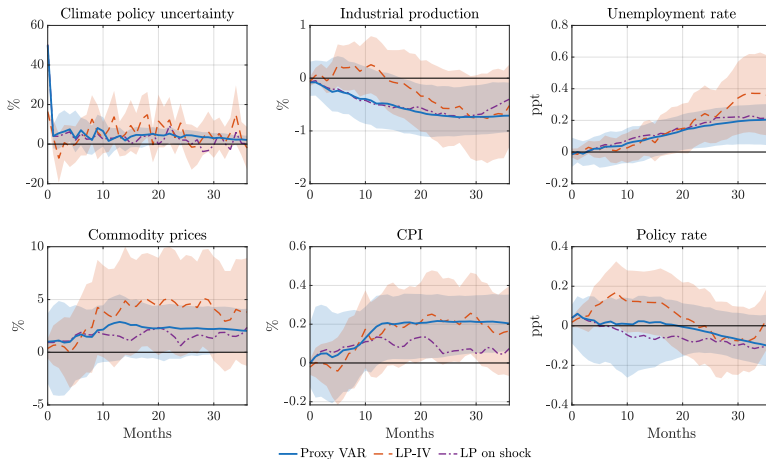
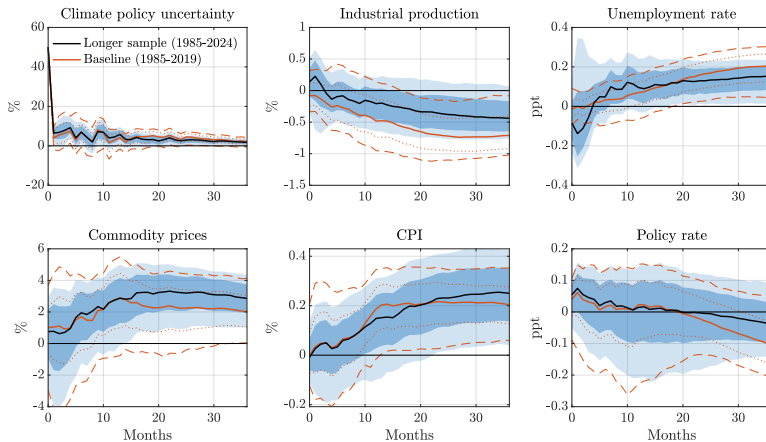
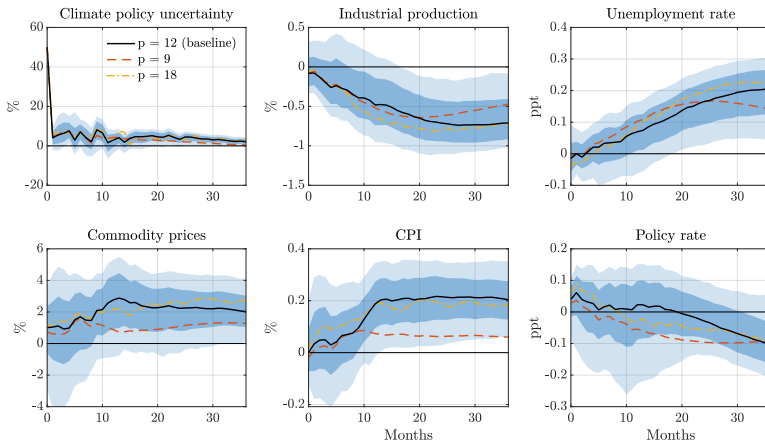


Figure 26: Sample period



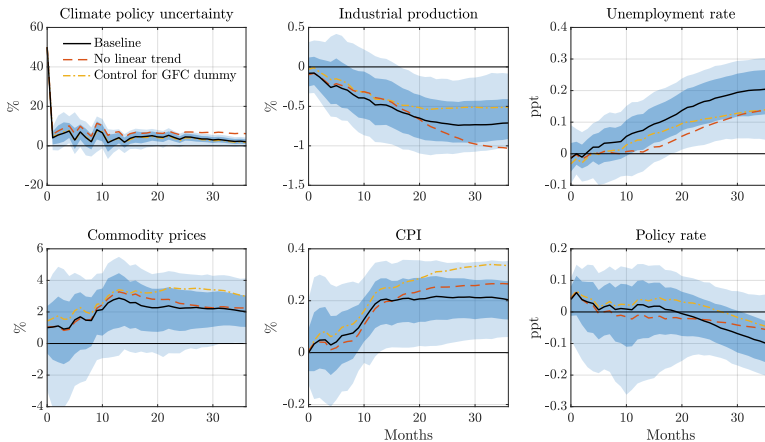
Relaxing VAR Assumptions

Figure 27: Lag Order



Relaxing VAR Assumptions

Figure 28: Deterministic Variables



Why is Climate Policy Uncertainty Inflationary?

- Response of prices to uncertainty shocks theoretically ambiguous
- Key channels:
 - **Precautionary demand:** Higher uncertainty leads agents to cut spending, **reducing prices** via lower demand
 - **Real options channel:** After a rise in uncertainty, firms delay investment and hiring
 - **Precautionary pricing:** Increased uncertainty raises the potential for higher future costs, leading firms to **raise prices** preemptively
- Price response depends on relative strength of supply- and demand-side effects

► Simple model

Simple Two-Sector NK Model

- Why are CPU shocks inflationary?
- Study propagation of different uncertainty shocks in NK model with two sectors:
 - **Energy sector** producing energy/emissions using labor
 - **Non-energy sector** producing consumption good using energy and labor
- Standard household sector and fiscal/monetary authority

Non-Energy Sector

- Technology:

$$y_t = Z_{x,t} e_t^\alpha n_{x,t}^{1-\alpha}$$

- Cost-minimization:

$$p_{e,t} = \alpha mc_t \frac{y_t}{e_t}$$

$$w_t = (1 - \alpha) mc_t \frac{y_t}{n_{x,t}}$$

- Price setting:

$$\pi_t (\pi_t - \bar{\pi}) = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \pi_{t+1} (\pi_{t+1} - \bar{\pi}) \frac{y_{t+1}}{y_t} \right] + \frac{\varepsilon}{\Omega_p} \left(mc_t - \frac{\varepsilon - 1}{\varepsilon} \right)$$

- Consider uncertainty shock about productivity, $\sigma_{Z_{x,t},t}$

Energy Sector

- Technology:

$$e_t = Z_{e,t} n_{e,t}$$

- Cost-minimization:

$$(1 - \tau_t) p_{e,t} = \frac{\eta}{\eta - 1} \frac{w_t}{Z_{e,t}}$$

- Price setting:

$$\pi_t^e (\pi_t^e - \bar{\pi}^e) = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \pi_{t+1}^e (\pi_{t+1}^e - \bar{\pi}^e) \frac{e_{t+1}}{e_t} \frac{p_{e,t+1}}{p_{e,t}} \right] + \frac{\eta}{\Omega_p^e} \left(\frac{w_t}{Z_{e,t} p_{e,t}} - \frac{(1 - \tau_t)(\eta - 1)}{\eta} \right)$$

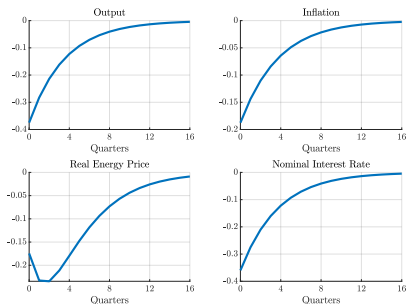
- In line with the data, assume that energy prices more flexible than goods prices

- Consider uncertainty shock about carbon tax, $\sigma_{\tau_t,t}$

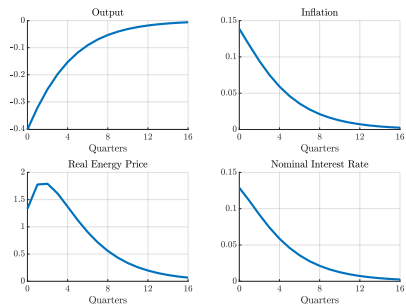
Differential Impact of Uncertainty Shocks

Figure 29: Uncertainty Shocks in Model

(a) Uncertainty about Productivity



(b) Uncertainty about Carbon Tax



Differential Impact of Uncertainty Shocks

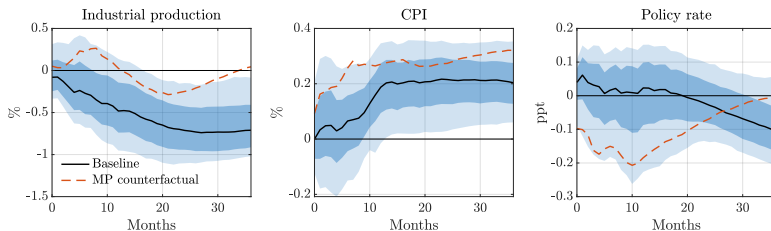
- Uncertainty about productivity in non-energy sector has very different implications
- Consistent with data, uncertainty about productivity is **disinflationary** while climate policy uncertainty is **inflationary**
- Precautionary pricing channel dominates precautionary demand channel for climate policy uncertainty

The Role of Monetary Policy

- How important is **monetary policy** for transmission of climate policy uncertainty?
- Perform a **counterfactual exercise** using McKay and Wolf (2023) approach
 - Use monetary shocks from Bauer and Swanson (2023)
 - Robust to Lucas critique
- Exercise: Impose same monetary reaction after CPU shock as for EPU shock

The Role of Monetary Policy

Figure 30: Monetary Policy Counterfactual



- Monetary policy response **matters** for the transmission of CPU shocks
 - Mitigates industrial production response substantially
 - Comes at cost of tolerating slightly higher inflation
- Should monetary policy respond **differently** to different sources of uncertainty?

Firm-Level Impacts

- Estimate effects on firm-level outcomes using panel local projections:

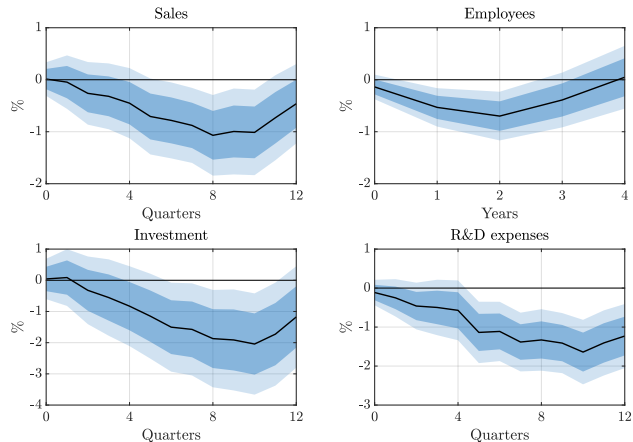
$$y_{i,t+h} = \mu_{i,h} + \beta_h \varepsilon_{1,t} + \gamma_h' \mathbf{x}_{i,t-1} + \nu_{i,t+h}$$

where $\varepsilon_{1,t}$ is the identified **climate policy uncertainty shock**

- **Outcomes:** Sales, employees, investment, R&D

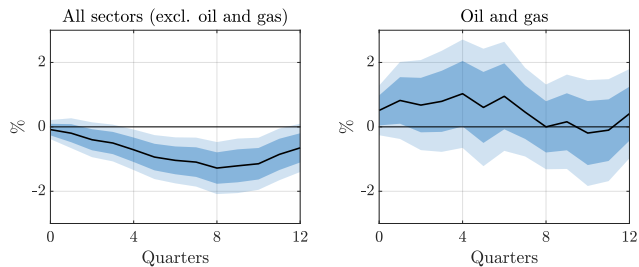
Average Effect

Figure 31: Average Effect on Firm Outcomes

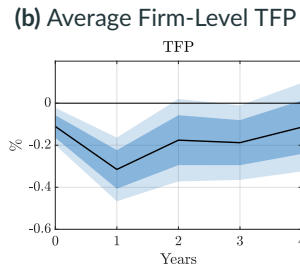
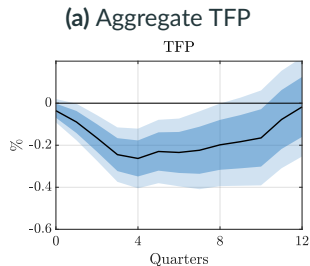


- **Sales and employees** fall significantly
- Substantial fall in firm-level **investment** and **R&D**
- Evidence consistent with macro results

Figure 32: Heterogeneous Effects by Sector



Longer-Term Impacts



- Climate policy uncertainty is a drag on investment and innovation
- Distorts allocation: Over-investment in firms with uncertain long-term viability